

Sketching with ImageNetion: A Framework for Creative Sketch Recognition and Generation

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Abstract

In human visual creativity, abstract figures often evoke multiple and imaginative associations, a capacity widely observed in art and design. We address the problem of how to generate drawings from degraded sketches that not only satisfy structural constraints but also exhibit multiple creative interpretations. The core challenges lie in the uncertainty of semantic recognition and the scarcity of multi-interpretation data. We propose ImageNetion, a framework for structure-constrained creative generation, which operates in two stages. In the static training stage, we adopt a supervised semantic alignment strategy to establish a basic correspondence between degraded sketches and textual descriptions. In the dynamic self-feedback training stage, we integrate a diffusion model equipped with structural control (ControlNet) and low-rank adaptation (LoRA), treating it as a constraint-aware prior of human intuition. Through a recognition-generation-discrimination mechanism, the recognition model can improve its multi-interpretation generation ability under unsupervised conditions, while alleviating the issue of data scarcity. Based on a synthetic sketch dataset covering more than 400 categories, our experiments demonstrate that the proposed method generates visually creative outputs with multiple interpretations under strict structural constraints, outperforming current mainstream approaches in both structural consistency score and top- k text-image consistency score. Furthermore, we design a user study to evaluate the method's superiority in terms of creative diversity and human preference.

Keywords: *Creative generation, Computational art, Sketch synthesis, Multimodal alignment*

1. Introduction

Amodal completion [17] is a perceptual mechanism extensively studied in cognitive science. It refers to the human ability to perceptually “complete” the structure of an ob-

ject even when the visual input is partially occluded or degraded. This capacity enables humans to recognize objects and scenes rapidly, even when only fragmentary information is available. As illustrated in Figure 1, when the body of a dog is obscured by a fence, observers still perceive it as a complete animal. Similarly, certain geometric shapes can be easily restored to their original forms even if portions are missing or corrupted. This perceptual completion mechanism relies not only on low-level visual features, but also on prior experience, semantic knowledge, and schema activation, and it plays a critical role in visual cognition.

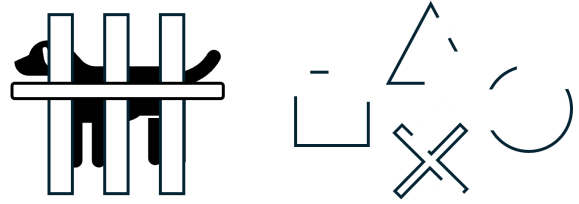


Figure 1. Example of amodal completion.

Inspired by this cognitive mechanism, we pose a new research question: given extremely sparse input, such as degraded sketches, is it possible not only to preserve the original information at the structural level but also, under structural constraints, to simulate the associative process of human imagination and generate creatively completed sketches? Unlike the large body of work on image inpainting or occlusion recovery, our concern is not about restoring the occluded part of a dense image to obtain a single ground-truth solution. Rather, we investigate how to construct visual content that is inherently multi-interpretable and creative, starting from abstract, open-ended strokes that lack contextual information.

Driven by advances in generative models, recent years have witnessed significant progress in image restoration and occlusion-based completion tasks [1, 32]. However, these methods typically rely on rich contextual information in the input. By contrast, sketches are not only extremely sparse but also lack a unified visual grammar.

Previous work has also examined sketch restoration [12, 20, 21, 27], but these studies primarily focus on reconstruct-

ing the original sketch category, producing deterministic outputs without faithfully recovering the conditional strokes in their spatial structure. In fact, when a sketch is heavily degraded, its corresponding semantic space becomes highly uncertain and open-ended. This implies that image generation guided by degraded sketches should not be viewed as a one-to-one information recovery problem, but rather as a process of semantic construction constrained by spatial structure.

To achieve multi-interpretation and creative sketch generation, two challenges must be addressed. First, if a supervised approach is adopted, a sufficiently large dataset of multi-interpretable sketches sharing common structural patterns is required. Such data, however, must be carefully designed by human creators and cannot be easily produced in bulk through computational methods, leading to a scarcity problem. Second, there exists a recognition-generation gap: how to extract diverse semantic information from highly degraded sketches and, based on this information, produce structurally consistent yet creative sketch completions.

To this end, we first construct a label set consisting of more than 400 concrete, structurally simple, and commonly used concepts. Based on this set, we employ a generative model to synthesize sketch images, and then apply controlled degradation strategies to create weakly semantic input samples that simulate highly abstract stroke structures. On this basis, we begin with the static training stage: a supervised semantic alignment strategy is adopted to enable BLIP-2 [11] to establish stable semantic associations between degraded sketches and their corresponding textual descriptions, thereby equipping the model with a fundamental ability to recognize abstract strokes. Subsequently, we introduce a diffusion-based generator [25] enhanced with ControlNet [36] and LoRA [7]. Leveraging its strong structural constraints and visual priors, the generator produces structurally consistent yet multi-interpretable sketches under appropriate textual prompts. Building on this generator, we design the second stage—a self-feedback training mechanism under unsupervised conditions. In this stage, the recognition model retrieves candidate semantics from the prompt library based on degraded sketches, the generator produces corresponding images, and the semantic-visual consistency between prompts and generated outputs is used as a feedback signal to refine the recognition model. This strategy not only enhances the capability for multi-interpretation generation but also effectively mitigates the scarcity of multi-interpretation data.

ImageNetic is capable of recognizing semantics from highly degraded abstract sketches and generating diverse images, thereby offering multiple possible completions for artistic creation. This ability not only expands the ideation space of illustration and concept design, but also serves as an educational tool that transforms abstraction into con-

creteness, stimulating children’s imagination and spatial perception. At the same time, the method supports pattern and visual symbol design by rapidly generating multi-style logos, decorative elements, or interface icons, thus providing creative support for graphic design and brand identity development.

Our contributions can be summarized as follows:

- We propose a two-stage training algorithm that addresses the novel problem of creative generation from degraded sketches, formulated within a recognition-generation framework.
- A synthetic sketch dataset grounded in concrete semantic labels is constructed to facilitate training for sparse sketch recognition.
- We introduce the structural consistency score and the top-k text-image consistency score as evaluation metrics to assess structure-constrained creative generation.
- A Human Evaluation involving 50 participants with backgrounds in digital media creativity is conducted to validate the quality and creativity of the outputs produced by ImageNetic.

2. Related Work

Amodal Completion. The task of amodal completion aims to infer the full appearance of objects under occlusion. Recent studies have increasingly combined recognition and generation strategies. One line of work explicitly models occlusion and predicts masks, inferring hidden structures from visible features [10, 35]. Another line introduces generative models as priors for appearance or structure, with diffusion models in particular being applied to reconstruct occluded regions from partial observations [18, 34]. A further direction emphasizes unsupervised or zero-shot capability within the recognition-generation loop, where frozen pretrained models enable amodal prediction without additional training [14]. Current research on amodal completion, however, primarily targets dense photographic images for occlusion recognition, segmentation, and prediction, rather than sparse and degraded sketches. Our work extends this cognitive concept toward the creative recognition and generation of sketches.

Sketch Generation. Traditional sketch synthesis methods typically rely on image processing techniques, such as extracting image edges to mimic sparse sketch lines [2, 31], or modeling sketch styles for style transfer [15]. With the rapid development of deep learning, significant advances have been achieved in this domain. Early approaches simulated human drawing sequences by modeling sketches as stroke offsets and state sequences [6], later incorporating

image encoders as conditions to improve category generalization [3]. Subsequent studies introduced Transformers for vector sketch modeling, either through large-scale self-supervised pretraining [13] or direct sequence encoding [24], which markedly improved the reconstruction and interpolation of complex sketches. More recently, pre-trained diffusion models have demonstrated strong sketch generation capabilities [19, 23]. Ke *et al.* [9] introduced a diffusion-guided network into the denoising process to produce raster sketches from photos, while Xing *et al.* [33] leveraged diffusion-based visual priors to guide Bézier curve optimization. Vinker *et al.* [29, 30] employed CLIP-based [22] semantic guidance to generate both object-level and scene-level vector sketches from photographs, enabling control over abstraction by reducing vector curves. From this perspective, our work can be viewed as a reverse process: starting from abstract, degraded sketches, we add strokes to construct meaningful and interpretable results.

Sketch Restoration. Sketch restoration is perhaps the most closely related direction to our work, where the goal is to recover a complete sketch from a damaged input. SketchHealer [27] first introduced this task using graph neural networks to encode sketch structures, and SketchHealer2.0 [21] extended it with perceptual loss from pre-trained CNNs and a dual-objective optimization strategy dynamically weighted by the missing rate. Other approaches avoid relying on vector or stroke order: Sketch-Lattice [20] converts pixel sketches into graph structures via lattice sampling, while Lmsr-pix2seq [12] reconstructs complete sketch sequences directly from raster input. These studies generally emphasize one-to-one sketch recovery. Some efforts, however, explore creative generation, such as SketchHealer2.0 combining latent vectors of two sketches to synthesize new ones, or Sketch-RNN [6] interpolating between sketches and completing strokes within a given category. While similar in spirit, these methods remain category-specific. In contrast, our work targets the plausible interpretation and conditional generation of weakly semantic strokes with cross-category diversity. To this end, we adopt raster sketches as both input and output, fully leveraging the structural priors embedded in diffusion models.

Motivation. Although prior studies have achieved remarkable progress in completion, restoration, and generation, they remain limited in addressing the task of multi-interpretation creative generation from sparse and semantically ambiguous abstract sketches. Image inpainting and amodal completion methods rely on dense contextual information, which allows them to infer missing content and structures deterministically, but makes them unsuitable for sparse sketch styles. Sketch restoration approaches generally aim to reconstruct ground-truth sketches of the same category, lacking cross-category semantic divergence and often suffering from structural inconsistency. Sketch gener-

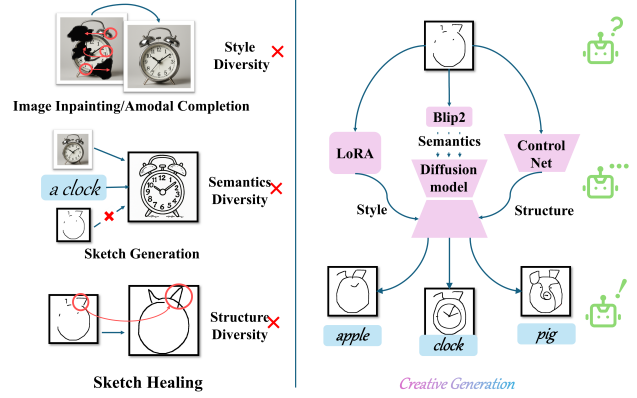


Figure 2. Existing problems and motivation.

ation and stylization methods, while effective under explicit semantic prompts, fall short when facing abstract strokes with little or no semantic content, as they lack mechanisms to extract and filter plausible semantics from structural cues.

To bridge this gap, we redefine stroke recognition and creative sketch generation along three dimensions—Style, Structure, and Semantics, as shown in Figure 2. Specifically, Sd-LoRA is used for lightweight style control, ensuring that outputs conform to sketch-like abstraction; ControlNet enforces stroke-level structural constraints, preserving the input configuration; BLIP-2 enables text-image retrieval to close the recognition-generation gap and introduce semantic diversity; and diffusion models serve as the generative backbone. Within this framework, the generator is not merely a rendering tool, but also functions as a prior aligned with human intuition under style and structure constraints. By serving as a self-feedback signal source, it provides consistency judgments that iteratively refine the recognition model, thereby enhancing multi-interpretation generation even under conditions of data scarcity.

3. Methodology

As illustrated in Figure 3, our framework begins with dataset construction. The generator is a ControlNet-based diffusion model enhanced with LoRA, which produces structurally consistent and semantically plausible sketches conditioned on degraded inputs. The BLIP-2 model is fine-tuned in two stages: static training equips it with basic recognition ability for sparse strokes, followed by generative self-feedback training that refines prompt selection and semantic alignment. Finally, evaluation metrics include the structural consistency score and the top-k text-image similarity, which jointly assess the structural fidelity and semantic plausibility of the generated results.

3.1. Dataset

Although existing hand-drawn sketch datasets are abundant [4, 5, 8], their quality is highly uneven. For instance,

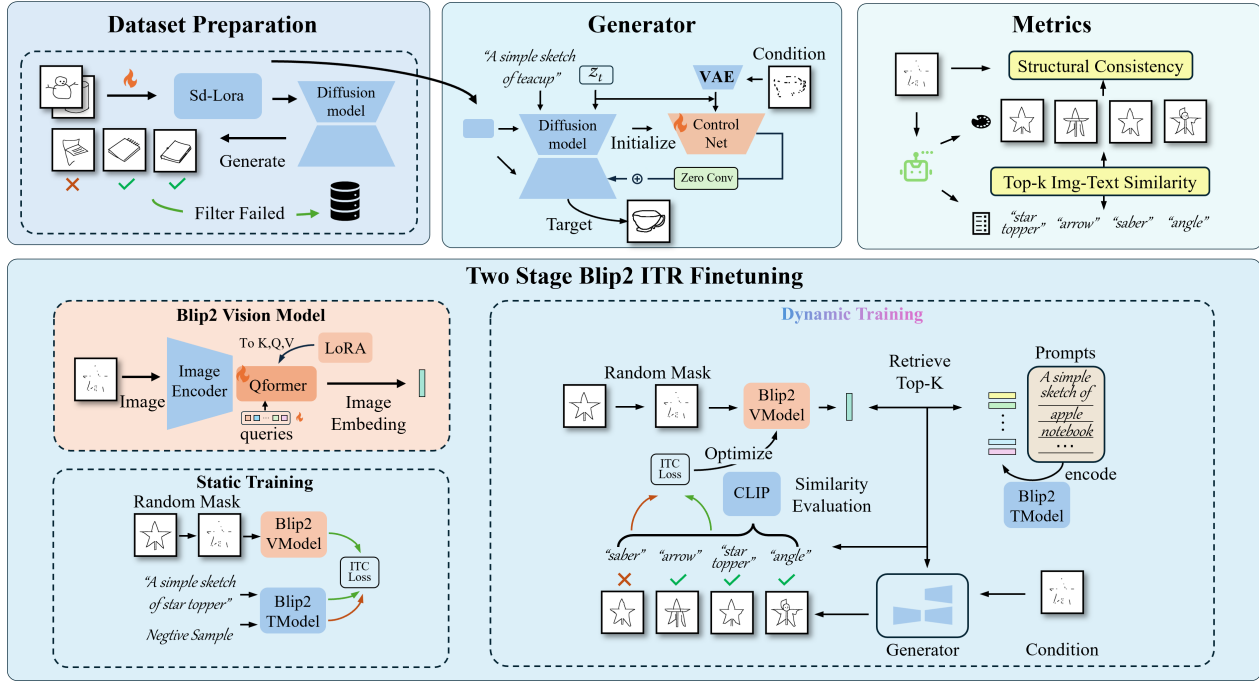


Figure 3. Overview of the ImageNetic pipeline.

the large-scale QuickDraw dataset contains many sketches that are difficult to recognize, with distorted shapes and disordered strokes. Moreover, the level of stroke complexity varies across hand-drawn datasets. Some sketches retain substantial information even after masking, making them unsuitable for training models on abstract structural association.

To address this, we extracted a small number of high-quality sketches from the work of Eitz *et al.* [4] and manually drew additional samples to construct a supplementary training set for sketch LoRA training based on diffusion models. We then employed a language model to generate a large set of object labels, focusing on items that are common in daily life and structurally simple. Example labels include apple, umbrella, Christmas tree, and sneaker. Each label was inserted into the prompt “a simple sketch of {label}” and fed into a sketch-capable diffusion model to generate 15 samples per label.

During generation, we used a CLIP model to evaluate the similarity between prompts and generated results, filtering out failed generations. Labels that consistently produced poor outputs were discarded, and semantically redundant categories were removed via word-embedding similarity filtering. The resulting synthetic sketch dataset comprises 448 categories, as shown in Figure 4.

3.2. Generator

The generator is built upon a diffusion model with sketch LoRA (Section 3.1) providing style constraints. Since the structural correspondence between degraded sparse strokes

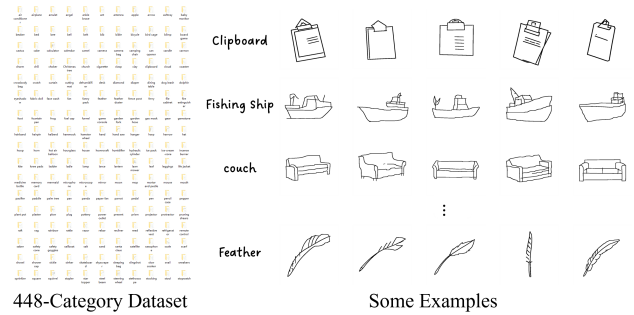


Figure 4. Synthetic sketch dataset constructed in this work. The synthetic sketches, selected for sketch LoRA training, feature simple lines and clear structures. Compared with human-drawn datasets, they offer better recognizability and are more suitable for recognition-generation training under degraded conditions.

and complete sketches is well suited to ControlNet, we adopt it to impose structural constraints. Using the hand-drawn sketch dataset [4], we randomly erase parts of sketches for degradation, with some further degraded into dot patterns to simulate varying levels of input sparsity. In this setting, degraded sketches serve as weakly semantic structural control signals, while the original sketches act as target outputs to form paired training data.

Considering that ControlNet always operates jointly with sketch LoRA during inference, the LoRA parameters are injected into the cross-attention layers of the diffusion model at training initialization and remain frozen throughout the entire ControlNet training process. To maintain parameter consistency between ControlNet and the backbone network,

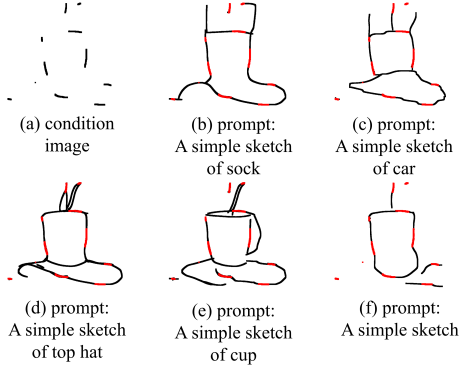


Figure 5. Generation results under manual prompts. We manually drew a conditional stroke sketch with a faint cylindrical structure and a flat base. When prompted with terms such as sock, cup, or hat, which share similar structures, the generated results aligned well with the prompts. In contrast, when using prompts like car, the outputs deviated significantly from the intended semantics.

when duplicating the UNet upsampling blocks for ControlNet initialization, the corresponding LoRA parameters are likewise injected.

We observe that, in the absence of meaningful prompts, the generator alone struggles to produce semantically clear sketches, although it effectively connects sparse strokes into coherent structures (Figure 5). Interestingly, when the input prompt refers to an object whose shape aligns with the sparse strokes, the generator consistently finds plausible, and sometimes remarkably clever interpretations. Conversely, when the prompt and the strokes are unrelated, the generated results are semantically meaningless. This observation indicates that the generator possesses a structural prior aligned with human intuition, which makes it suitable as a component of the self-feedback recognition training.

3.3. Two-stage Training

In this work, we employ the BLIP-2 vision model [11] as the retrieval backbone for degraded sketch-text alignment, owing to its strong multimodal representation capacity and the compatibility of its Q-Former module with low-rank adaptation. To this end, we design a two-stage training process to progressively enhance the model’s ability to interpret weakly semantic sketches and align them with textual prompts. During training, LoRA parameters are injected into the Key, Value, and Query components of the Q-Former’s attention layers, while the Query Tokens fed into the Q-Former are simultaneously fine-tuned. These Query Tokens are responsible for extracting visual information relevant to downstream tasks. The text embeddings, in contrast, are provided by the frozen BLIP-2 text model throughout training.

Static Training Stage. In this stage, we simulate common cases of missing or blurred strokes in human sketches by randomly erasing parts of strokes from the synthetic

dataset. These degraded sketches are fed into the vision encoder of BLIP-2 to obtain the corresponding image embedding vector. We then apply a text-image contrastive loss: the degraded sketch embedding S and its paired textual description t^+ are treated as a positive pair, while N randomly sampled alternative descriptions t_i^- serve as negative samples. Cosine similarities between S and t^+ and between S and t_i^- are computed, yielding ℓ^+ and ℓ_i^- , shown in Eq. 1 and Eq. 2 respectively. The final static loss function $\mathcal{L}_{\text{static}}$ is formulated in an InfoNCE manner as shown in Eq. 3.

$$\ell^+ = \text{sim}(S, t^+) \quad (1)$$

$$\ell_i^- = \text{sim}(S, t_i^-) \quad (i = 1, \dots, N) \quad (2)$$

$$\mathcal{L}_{\text{static}} = -\ell^+ + \log \left(\exp(\ell^+) + \sum_{i=1}^N \exp(\ell_i^-) \right) \quad (3)$$

Dynamic Self-feedback Training Stage. In this stage, the parameters from the static stage are used for initialization, which provides the model with a basic capability for recognizing degraded inputs. The purpose of this stage is to further improve the accuracy of top- k prompt retrieval through collaboration with the generator. Here, we define “accuracy” as the ability of the retrieved prompts to guide the generator to produce images that not only conform to the structure of the degraded sketches but also exhibit high semantic consistency with the prompts and maintain high visual quality.

During training, we again apply random erasing to the synthetic sketches and extract image embeddings using BLIP-2. These embeddings are then used to retrieve the top- k most similar prompts from the prompt library. The retrieved prompts are passed into the generator described in Section 3.2, which leverages its built-in visual priors, with the degraded sketch provided as the conditional input to ControlNet. To evaluate the consistency between prompts and generated results, we employ a frozen CLIP model to measure the alignment between each image-text pair. Prompts showing higher alignment are collected as the positive set $\{t_j^+\}_{j=1}^P$, while those with lower alignment form the negative set $\{t_i^-\}_{i=1}^N$.

When computing ℓ_j^+ for dynamic training, we assign softmax weights to each ℓ_j^+ based on its CLIP similarity score m_j , with temperature parameter λ , as formulated in Eq. 4. The weighted score is then used to compute the final ℓ^+ in Eq. 5. The definitions of ℓ_i^- and the dynamic loss $\mathcal{L}_{\text{dynamic}}$ remain in the same form as Eq. 1 and Eq. 3 in the static stage.

$$\ell_j^+ = \text{sim}(S, t_j^+), \quad w_j = \frac{\exp(m_j/\lambda)}{\sum_{u=1}^P \exp(m_u/\lambda)} \quad (4)$$

$$\ell^+ = \sum_{j=1}^P w_j \ell_j^+ \quad (5)$$

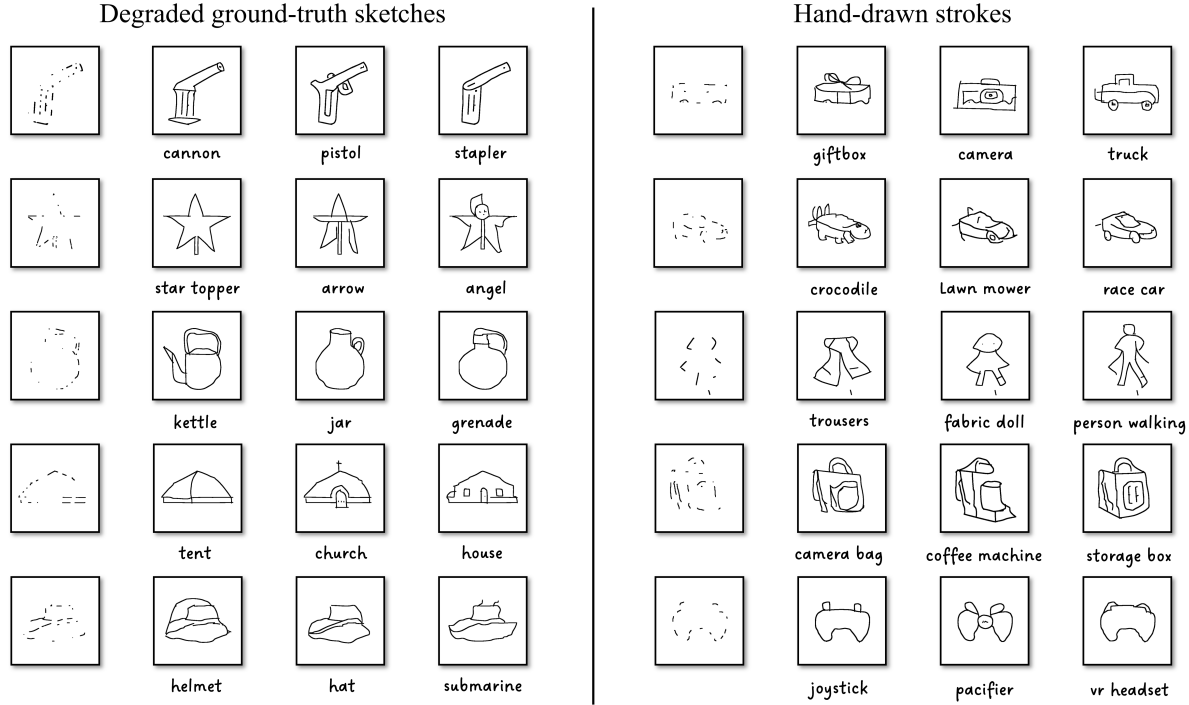


Figure 6. Creative generation results from text-image retrieval on two datasets.

4. Results and Analysis

4.1. Setting

Implementation. All experiments were conducted on a single NVIDIA 4070 GPU. We set top- k to 6, and the generator employed in training was based on Stable Diffusion 1.5. During the generation process, we used the DPM-Solver++ [16] scheduler with 7 inference steps to reduce training time. The static training phase was carried out on a dataset of approximately 7k samples for 10 epochs. The dynamic training phase used the same dataset and was run for 10k steps. Training was performed with the Adam optimizer. We set the learning rate to 5e-6 and adopted a cosine learning rate schedule.

Data Preparation. We prepared three datasets. DS1 is the synthetic sketch dataset constructed in this work, which was used for both visual recognition training and testing. DS2 consists of hand-drawn strokes that carry no explicit semantic meaning but exhibit certain structural patterns; it was employed to evaluate the model’s recognition and generation capability in the absence of ground-truth sketches. DS3 corresponds to the dataset used by SketchHealer and Lmsr-pix2seq, comprising 17 categories selected from the QuickDraw [8] dataset, and was used for comparative evaluation.

Evaluation Metrics. Evaluating the results of diverse generation from degraded sketches is challenging. We therefore propose the following metrics. First, the pixel

structures provided by the input sketch should also appear in the generated image, which we define as the *structure score*. This score focuses on the set of effective foreground pixels, denoted as

$$C = \{p \mid \mathcal{C}(p) = 1\}, \quad G = \{q \mid \mathcal{G}(q) = 1\},$$

where C represents the set of foreground pixels in the conditional sketch, and G represents the set of foreground pixels in the generated image. We then compute the recall of C within G .

Let $d_G(p)$ be defined as:

$$d_G(p) = \min_{q \in G} \|p - q\|_2, \quad (6)$$

that is, the minimum Euclidean distance between a foreground pixel $p \in C$ and any foreground pixel $q \in G$. Given a distance tolerance threshold t , if $d_G(p) < t$, the pixel p is counted as correct; otherwise, it is assigned zero. The proportion of correct pixels over all pixels in C yields the structure score as follows:

$$S_{\text{struct}} = \frac{1}{|C|} \sum_{p \in C} \mathbb{I}[d_G(p) < t]. \quad (7)$$

For a degraded sketch, a multi-interpretation generation system should produce a set of candidate explanations and corresponding conditional generations. In this work, we select the top- k text-image retrieval results $T = \{t_j\}_{j=1}^k$, and use the generator to obtain a set of generated images

$I = \{p_i\}_{i=1}^k$. We then compute the average similarity between the top- k explanations and their generated results using the SigLIP2 [28] model, as defined in Eq. 8. The metric $\text{SigLIP}_{\text{top-}k}$ is employed to evaluate the overall plausibility of the k text-image explanations proposed by the system for a given degraded sketch:

$$\text{SigLIP}_{\text{top-}k} = \frac{1}{k} \sum_{i=1}^k \text{sim}(t_i, p_i). \quad (8)$$

Finally, we employed the LAION-AI Aesthetic Predictor [26] to assess the aesthetic quality of the generated sketches. Since this predictor was not specifically designed for sketches, we further incorporated an additional aesthetic quality evaluation in the human study, thereby obtaining results that are more closely aligned with human preferences.

4.2. Main Results

As the degree of degradation increases, our system is able to generate results with greater diversity. Conversely, as degradation decreases, the SigLIP scores of multi-prompt retrieval also decline, since the semantic content of the sketch becomes more tightly bound to its spatial structure. Representative examples under higher degradation are shown in Figure 6. On the left, where the input consists only of fragmented strokes derived from a degraded ground-truth sketch, the system interprets a triangular-like shape as a tent, a church, or a house, while preserving the essential contours and local details in the generated results. Even for categories with similar shapes but divergent semantics—such as grenades and jars—the model can still generate structure-complete results that are consistent with the given prompts. On the right, where no ground-truth sketch constraints are available, the system is still able to reasonably interpret loosely drawn strokes as objects such as a gift box, a car, or a human figure. In these cases, the generation process effectively leverages the distribution of the original strokes to achieve a balance between structural fidelity and semantic plausibility.

Ablation. We conducted ablation studies on the unmodified BLIP-2 model, Ours-stage1, and Ours-stage2 to examine the differences in generation quality across training stages. Table 1 reports the average SigLIP scores for the top- k results, which are computed by Eq. 8. For DS1, the unmodified BLIP-2 used as a visual encoder yields relatively low SigLIP scores, and its retrieved prompts are mostly concentrated within a small number of categories, indicating limited discriminative ability on degraded sketches. After the first-stage static training, the average top- k SigLIP scores show a clear improvement. When further refined with the second-stage dynamic self-feedback training, the top- k SigLIP scores consistently surpass those from stage one. The overall trend on DS2 is similar to that

Table 1. Ablation results on the two datasets. top- k denotes the average SigLIP score over the first k retrieved text-image pairs.

top- k	DS1			DS2		
	Blip2	Ours-stg1	Ours-stg2	Blip2	Ours-stg1	Ours-stg2
1	44.8	72.2	78.5	39.7	63.8	70.1
2	44.4	64.8	72.2	31.4	56.2	67.8
3	40.1	61.0	66.1	29.9	52.8	64.2
4	38.7	58.3	62.9	29.2	48.3	60.0
5	37.5	56.5	61.6	28.0	46.4	58.1
6	37.2	54.7	59.6	26.6	44.0	57.1
7	37.0	52.8	58.1	25.8	42.2	56.4
8	36.7	51.1	56.6	25.2	40.6	54.2

on DS1, but since the inputs are not derived from degraded ground-truth sketches, all three models exhibit reduced performance. In this scenario, the performance gap between Ours-stage2 and Ours-stage1 becomes more pronounced, suggesting that the self-feedback training enables the model to better adapt to more challenging inputs.

4.3. Baseline Comparison

Although the goal of ImageNetion is not sketch repair, its task setting is closely related. Based on this, we select two representative sketch repair methods, SketchHealer [27] and Lmsr-pix2seq [12], as baselines to examine the characteristics of our approach in degraded-sketch conditional generation compared with them.

For a fair comparison, our method was trained for text-image retrieval on the same set of categories as the baselines. SketchHealer proposed a graph-based sketch restoration approach, which models patch images of sketches and their adjacency relations using a graph convolutional network, and optimizes the generation process with a reconstruction loss. Lmsr-pix2seq, on the other hand, regresses sketch sequences in a language modeling framework, combined with a masking strategy to achieve robust sketch completion. For comparability, we reproduced and evaluated these methods under the same datasets and occlusion ratios (p_{mask}). Since both SketchHealer and Lmsr-pix2seq generate sketch restorations in an end-to-end manner, their outputs do not provide explicit category interpretations. Therefore, when computing the SigLIP score, we selected the result with the highest similarity to the dataset category as the baseline interpretation. To ensure consistency, our method was also evaluated using only the top-1 interpretation.

Qualitative Evaluation. Figure 7 presents several restoration results from the baseline methods and our approach. When the masking ratio (p_{mask}) is relatively low, the baseline methods are able to produce reasonably good outputs, but structural deviations are already observable. As p_{mask} increases, SketchHealer begins to generate dense, vortex-like strokes, a phenomenon that also appears in Lmsr-pix2seq. At the same time, Lmsr-pix2seq often pro-

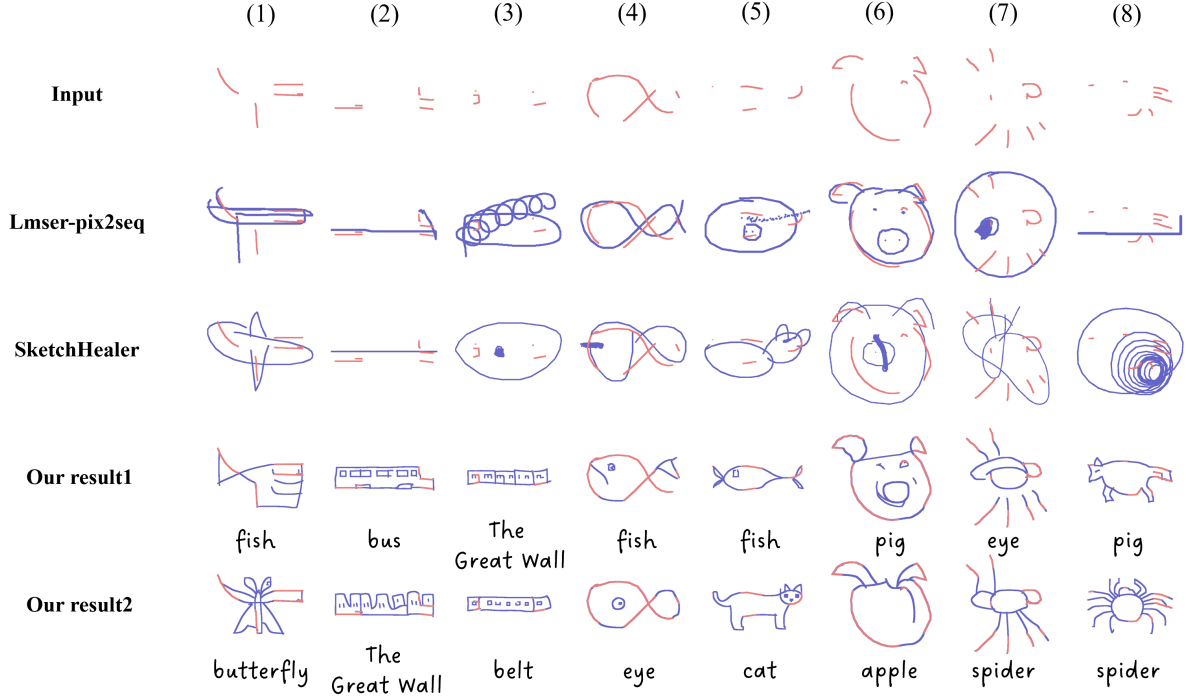


Figure 7. Comparison with baseline results: the red lines indicate the position of the input degraded sketches. Our results accurately align with the stroke traces and further generate outputs consistent with the prompts retrieved by the recognizer. We observe that in case (8), when the spider sketch is heavily degraded, the baseline methods produce shapes resembling eyes or butterflies, which are largely unrelated to the masked input. Similar issues can also be found in cases (5), (7), and (10).

duces relatively sparse and meaningless point patterns. In contrast, our method consistently preserves higher visual quality. In terms of structural consistency, Lmsr-pix2seq demonstrates somewhat better fidelity under mild degradation, but as p_{mask} increases, both baselines deviate significantly from the conditional sketches. We further observe that in some cases, when the input strokes are of particularly poor quality, both baseline methods tend to generate unrelated sketches. Finally, although only the top-1 generated structure is considered in the quantitative comparison, we also include some multi-interpretation results produced by ImageNetion, which reveal interesting phenomena of structural sharing.

Quantitative Evaluation. Table 2 reports the evaluation scores of the baseline methods and our approach under different degradation levels (p_{mask}). The structural scores are consistent with the qualitative observations, where the baselines exhibit significant structural drift, while our method maintains high structural consistency owing to the pixel-level control provided by ControlNet. In terms of the LAP aesthetic metric, our results outperform those of the baselines across all settings, benefiting from the high-quality image generation capacity of diffusion models. Regarding the SigLIP scores, when p_{mask} is low, the gap between our method and SketchHealer is relatively small. This is partly because the baseline scores are computed

Table 2. Comparison with baseline methods. The experiments are conducted on the baseline datasets. LAP refers to the LAION-AI Aesthetic Predictor metric. Struct and SigLIP are computed using Eq. 7 and Eq. 8, respectively. The best results are highlighted in bold.

Methods	Struct	LAP	SigLIP
SketchHealer ($p_{\text{mask}} = 0.5$)	0.23	3.85	59.61
Lmsr-pix2seq ($p_{\text{mask}} = 0.5$)	0.40	3.68	49.58
Ours ($p_{\text{mask}} = 0.5$)	0.99	4.11	62.56
SketchHealer ($p_{\text{mask}} = 0.6$)	0.24	3.82	61.43
Lmsr-pix2seq ($p_{\text{mask}} = 0.6$)	0.39	3.67	45.23
Ours ($p_{\text{mask}} = 0.6$)	0.99	4.14	65.41
SketchHealer ($p_{\text{mask}} = 0.7$)	0.20	3.83	55.45
Lmsr-pix2seq ($p_{\text{mask}} = 0.7$)	0.44	3.67	44.42
Ours ($p_{\text{mask}} = 0.7$)	0.99	4.15	66.94

by selecting the best-matching result by default, which favors them. More importantly, our method must generate sketch outputs that adhere to structural constraints while also aligning with the retrieved interpretations, a requirement that can affect visual quality. However, as p_{mask} increases, the SigLIP scores of the baseline methods decline sharply, whereas our method exhibits an upward trend. This occurs because with fewer residual strokes, the structural

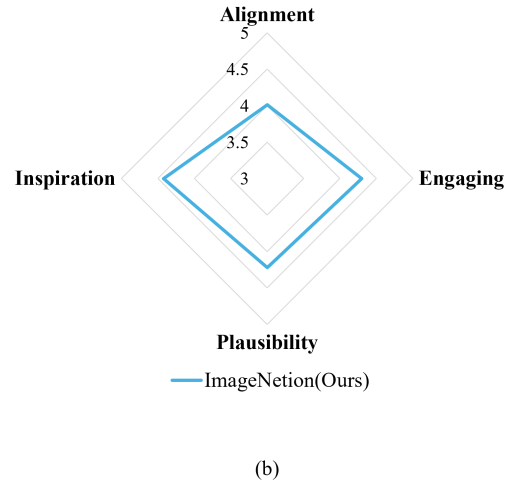
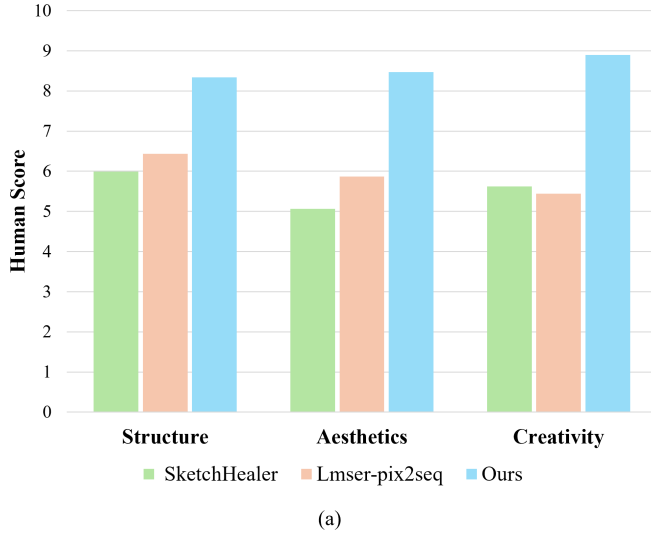


Figure 8. Baseline Human Evaluation. Figure (a) shows the human ratings of our method compared with SketchHealer and Lmser-pix2seq across three dimensions. Figure (b) presents the participants’ ratings of ImageNtion itself across four dimensions.

constraints become weaker, thereby allowing greater generative freedom. Since our framework is designed specifically for highly degraded sparse sketches, we set p_{mask} to start from 0.5.

4.4. Human Evaluation

Since the process of sketch interpretation and creative generation cannot be fully assessed by computational metrics, we conducted a human evaluation study involving 50 participants with backgrounds in digital media creativity, in order to obtain results more closely aligned with human preferences.

The study consisted of two parts. In the baseline comparison, participants were asked to rate the outputs of three methods across three dimensions: (1) the preservation of original strokes—whether the structural form and spatial positions of the input strokes appeared in the generated results; (2) aesthetic value; and (3) creativity. As shown in Figure 8 (a), our method significantly outperformed the baselines in all three dimensions, which is consistent with the quantitative trends reported in Section 4.3.

In the self-evaluation, participants were first shown the original abstract stroke sketches and asked to imagine possible interpretations. We then presented the set of text-image explanations generated by ImageNtion based on the same input. Participants evaluated the results along four dimensions: (1) alignment—the extent to which the generated outputs partially matched their own imagined interpretations; (2) engaging—the aesthetic appeal and engaging quality of the results; (3) rationality—the plausibility of the generated outputs as creative drawings derived from the original strokes; and (4) inspiration—the degree to which the results stimulated participants’ creative associations. As il-

lustrated in Figure 8 (b), our method achieved high scores across all four dimensions, with particularly strong recognition in engaging and inspiration. These findings indicate that our approach can deliver not only visually pleasing and engaging experiences, but also creative inspiration from abstract sketches, highlighting its potential for applications in design, education, and related fields.

5. Applications and Limitations

ImageNtion demonstrates distinct advantages in the recognition and multi-interpretation generation of highly degraded sketches. It not only provides multiple plausible concretized interpretations for the same sparse structure, but also dynamically presents progressively enriched results as additional strokes are added. Such capability can support rapid idea exploration in the early stages of artistic creation, enabling creators to obtain diverse visual solutions at varying levels of detail. In educational and training contexts, it can serve as an interactive tool that guides students to experience the creative evolution from abstract to concrete sketches, thereby fostering perceptual awareness of form and divergent thinking. In visual design, ImageNtion can assist in rapidly producing stylistically diverse yet structurally consistent variants of logos, icons, or decorative patterns, offering efficient creative support for interface and brand design. Figure 9 illustrates several potential applications of our framework.

While ImageNtion is capable of generating diverse associative interpretations from sparse strokes, several limitations remain. The current framework relies on retrieval from an existing textual corpus rather than open-ended text generation, which constrains linguistic diversity. It also focuses on object-level imagination and has not yet extended

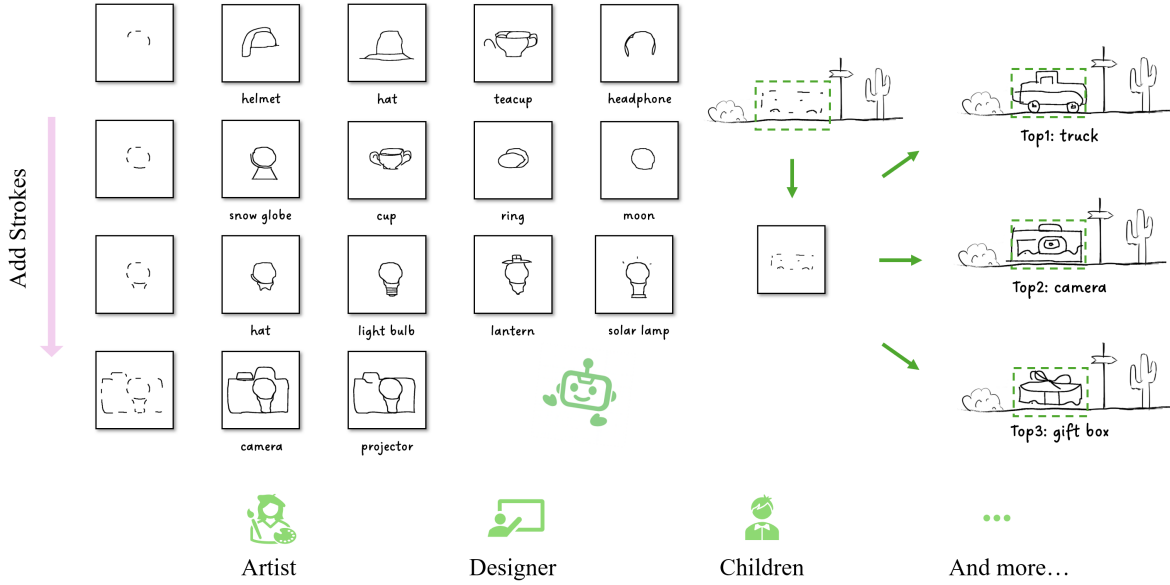


Figure 9. With the progressive addition of strokes, ImageNtion provides diverse interpretations. These interpretations can also be matched with existing scenes, allowing users to select suitable directions for depiction.

to scene-level inference from highly abstract inputs. In addition, the model sometimes tends to align inputs with structurally similar categories, reflecting a gap from the more free-form imaginative flexibility of human perception.

6. Conclusion

This study represents an exploratory attempt to integrate human perceptual imagination with the expressive capabilities of generative AI. We proposed the ImageNtion framework, whose core algorithm leverages the structural priors of diffusion models as feedback signals for the semantic recognition module. Under a cooperative optimization scheme, the framework enables the retrieval of multiple semantic interpretations from highly sparse strokes or degraded sketches, and generates creative outputs accordingly. The framework maintains strong stability even under severe degradation, surpassing baseline methods in both structural fidelity and semantic plausibility when evaluated on single-result metrics, while also achieving the highest ratings in human preference studies. Furthermore, the human evaluation indicates that most participants found the multi-interpretation outputs of our framework to be inspiring and engaging. These results suggest that ImageNtion holds promise for applications in artistic design and educational guidance for children’s cognitive development, and further extends the potential of sketches as a medium for cross-modal expression.

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